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Contents

S. No	Title	Page No.
1	Synthesis of silver nanoparticles of some edible basidiomycetes mushroom fungi using response surface methodology and its potential biological application **R Madhanraj, M Eyini and P Balaji**	01
2	Impact of CO_2 on growth, pigments yield and biochemical composition of marine microalga *Dunaliella salina* **A Shenbaga Devi, P Santhanam, S Jeyanthi, B Balaji Prasath and S Dinesh Kumar**	13
3	Fumaronitrile mediated cytochrome P450 (CYP) isoforms biotransformation enzymes responses in *Oreochromis mossambicus* **K Chinnadurai, M Eyini and P Balaji**	23
4	HPLC and biochemical techniques for secondary metabolites in *Garcinia indica* Choisy (Kokum) from transitional zones of Karnataka **Lingappa Sivakumar and Thirugnanasambandam Somasundaram**	35
5	Primary productivity of river chaliyar of Calicut district, Kerala, India **B Dhanalakshmi and P Priyatharsini**	48
6	Anti-bacterial activity, anti-inflammatory and anti- arthritic studies on mangroves by using in vitro model systems **M Babu Selvam and S Abideen**	54
7	Parasitic isopods of the family Cymothoidae from Indian fishes **S Ravichandran and G Ramesh Kumar**	65
8	Isolation and identification of pathogenic bacteria and its antibacterial susceptibility analysis in edible fish *Catla catla* **Mayavan Karthika, Shameem Shabana, Shamoon Muhasin and Venkatachalam Ramasubramanian**	72

9	Biogenic synthesis of silver nanoparticles from *Cardiospermum halicacabum* decorated with Graphene oxide for enhancing antibacterial ability **Gurusamy Sivaprakash, Gujuluva Hari Dinesh, Kulanthaisamy Mohan Rasu, Manoharan Dhivya and Alagarsamy Arun**	80
10	Studies on biosynthesis of xanthan gum using *Xanthomonas* sp., isolated from infected cotton leaves **V Ananthi and A Arun**	88
11	Characterization and determination of antibacterial activity of bacteriocin producing Lactic acid bacteria isolated from curd sample **V Ananthi and A Arun**	95
12	Antibacterial and immunostimulant influence of herbal extracts in grouper *Epinephelus tauvina* experimental culture against *Vibrio harveyi* Infection **T Citarasu, M Michael Babu and SMJ Punitha**	103
13	Assessment of bacteriological quality and presence of antibiotic resistant bacteria in vended sachet-packaged drinking water: potential threat of transmission of enteric pathogens and implications for public health **K Ramamoorthy and Clara G Sargunar**	117
14	Synthesis of chitin form shrimp dispel and its antibacterial activity **P Raja Rajeswari, R Shyamala Gowri, P Meenambigai and K Rajeswari**	132
15	Assessment of antibacterial activity of different solvent extracts of medicinal plant: *Aegle marmelos* **R Shyamala Gowri, R Vijayaraghavan, P Meenambigai and P Raja Rajeswari**	138
16	Effect of aqueous methanolic extract of *Tridax procumbens* on nonspecific immune response of fresh water fish **S Chinniah, T Sangeetha and Subeena Begum**	145
17	A study on biologically synthesize silver nanoparticles using red seaweed *Gracilaria gracilis* **V Veeramanikandan, PT Usha and P Balaji**	154

Impact of CO_2 on growth, pigments yield and biochemical composition of marine microalga *Dunaliella salina*

A Shenbaga Devi, **P Santhanam***, **S Jeyanthi, B Balaji Prasath** and **S Dinesh Kumar**
Department of Marine Science, School of Marine Sciences,
Bharathidasan University, Tiruhcirappalli, Tamil Nadu, India

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Abstract The present study was carried out to know the response of ocean acidification on the growth, pigments and biochemical's production of *Dunaliella salina*. Ocean acidification is one of the most well-known ocean perturbations. The increased growth has been observed in *D. salina* with increased CO_2 activity. However, the chlorophyll 'a', chlorophyll 'b' and phaeopigments concentration was found to be low in CO_2 fertilized group than the control group. The biochemical profile of *D. salina* showed high protein (3.02 µg/ml) and carbohydrate (0.41 µg/ml) while low lipid (10%) in acidified medium when compared to control group. Thus, atmospheric CO_2 increase could potentially promote *D. salina* growth and protein and carbohydrate in the acidified cultures were capable of compensating the pH change and they survived showing high productive results.

Key Words Ocean Acidification, Green algae, *Dunaliella saliina*, Atmosperic CO_2

Introduction

Global warming due to increased carbon dioxide concentration in the atmosphere is receiving a great deal of attention. Atmospheric increases of carbon dioxide are positively correlated with the amount of fossil fuels being burned (Stepan *et al*., 2001**)**. In an effort to retard this increase and, therefore, the greenhouse effect, most industrialized countries have joined in a policy to hold carbon dioxide emissions (Hileman, 1992). However, it is not clear if technology exists to achieve this goal. Atmospheric CO_2 enrichment would also result in a reduction of the pH and an increase in the CO_2 concentration of surface waters (Stumm and Morgan, 1996). In addition to climate change, another symptom of increased carbon dioxide (CO_2) levels in the atmosphere is ocean acidification. About a quarter of the CO_2 emitted into the atmosphere is absorbed by the ocean, and another quarter is absorbed by the natural environment on land. In water, the CO_2 forms a weak carbonic acid, making the oceans more acidic. Carbon dioxide plays a natural role in defining the pH of seawater (a brief account of measures of acidity such as pH, and the acid–base chemistry of the CO_2 - carbonate system in the oceans). CO_2 is the major and often limiting substrate for photosynthetic carbon assimilation in plants and other photosynthetic organisms.

Algae of the genus *Dunaliella* was among the most studied microalgae for mass culture. They are grown as a food source in aquaculture and for the production of high value protein and other bioactive constituents. More recently, these microalgae have been grown under certain conditions, e.g., nutrient stress, high light intensity, high salt concentration and pH extremes that induce the accumulation of highly important antioxidants (Matsukawa *et al*., 2000; Ip and Chen 2005).

*sanplankton@yahoo.co.in

For example, *Dunaliella* can produce and accumulate large amounts of commercially important compounds such as β-carotene and glycerol in response to stress conditions.

Dunaliella is a halophilic marine green alga found in environments with high salt concentrations (Oren, 2005). *Dunaliella salina* is found in high salinity environments such as salted brines, salt evaporation ponds, and hyper saline lakes (Brock, 1975). *Dunaliella salina* has adapted to survive in high salinity environments by accumulating glycerol to balance osmotic pressure. *D. salina* is also adapted to solar radiation using β-carotene to protect against ionizing energy. This combination offers potential in biotechnological applications for the purpose of commercial products such as lipstick due to β-carotene production (Schilipaulis, 1991). However, no study so far attempted to find out the effect of CO2 induced acidification on the growth and biochemical profile of *D. salina*. Hence the present experiment was made.

Materials and methods

Isolation and culture of *Dunaliella salina*

Microalgal samples were collected from salt pan located at Tuticorin (Lat 8.77^o N and Long. 78.14^o E), Tamil Nadu, Southeast coast of India using phytoplankton net of 25µm. The collected samples were immediately transported to the laboratory. At laboratory *D. salina* was isolated using serial dilution technique and cultivated using Conway's medium enriched with seawater (Walne, 1966). The seawater used for the microalgae culture was sterile, filtered and enriched with the full set of nutrients include macro and micronutrients, trace elements and vitamins and stored in a cool and dark place till use. All algae in this experiment were incubated in sub samples of this medium. Replicate cultures were grown and kept at 25°C in a thermostatically controlled room, illuminated with white fluorescence lamps with 12:12 h of light/dark regime. The temperature and salinity was maintained in the range of 23-25°C and 28-30 parts per thousand (hereafter ppt) respectively.

Experimental set up and CO_2 bubbling

The following treatment series were maintained for the CO_2 acidification studies on *Dunaliella salina*

Treatment 1: Control (No CO_2 acidification)-flasks covered with a cotton wad.

Treatment 2: CO_2 acidification-flasks covered with a cotton wad made airtight by wrapping it in a transparent film with a straw inserted in the middle for CO_2 acidification by blowing.

CO_2 bubbling was performed in the treatment 2, with the air blown into the culture for 2 minutes at four times a day.

Analytical procedures

The experiment typically lasted for 2 weeks. Data was collected every 3 days once for Chlorophyll and algal cell density. The pH was measured using an ELICO grip pH meter, Chlorophyll 'a' and 'b' was estimated by the standard method of Strickland and Parsons (1972). For growth analysis, 10 ml of algal culture was sampled from each replicates and measured by checking the absorbance at 660 nm using (1800 Shimadzu UV).

Estimation of biochemical profile

A known volume of cultures of *D. salina* species were harvested at the mid log phase by centrifugation using centrifuge at 6000 rpm for 10 minutes. The cells were oven dried at 60°C for 60 minutes and stored for

further biochemical analysis. The total lipid, carbohydrate and protein contents were quantitatively analyzed according to Folch *et al*., (1957), Dubois *et al*., (1956) and Lowry *et al*., (1951) respectively.

Analysis of chlorophyll

The biomass of *D. salina* was estimated by the standard method of Strickland and Parsons (1972). 10 ml of algal culture sample was filtered by using Millipore filtering system fitted with a 4.5cm diameter GF/C filter paper by applying low suction. Before filtering the sample, a thin bed of magnesium carbonate (2 ml) was made for effective filtration. After the filtration, the filter paper was removed by using clean forceps and grind with 90% acetone using mortar and pestle. The ground samples were transferred to screw cap test tubes and covered by using black cloth and incubated in the refrigerator for 24 hrs. After 24 hours incubation, the contents were regrinding with 90% acetone and centrifuged at 3000 rpm for 10 minutes. The optical density was measured at different wavelengths of 630, 645 and 665nm for chlorophyll 'a' estimation.

Analysis of phaeopigments

For the estimation of phaeopigments in the *D. salina* the method adopted for estimating chlorophyll 'a' is the same as described for phaeopigments. Further, the two drops of diluted hydrochloric acid was added to the cuvette then the samples well mixed and absorbance was measured at 665 and 750 nm. The amount of phaeopigments present in algae was estimated by subtracting the value of 750 nm from the value of corresponding 665 nm extinction using standard formula.

Statistical analysis

The data were expressed as the mean values ± standard deviation (SD). Statistical correlation tests were employed to make statistically significant conclusions about the data using SPSS Package ver. 14.0

Results

pH changes in the medium

In the CO_2 acidification study of *D. salina*, the optimum pH in control was 7.96 ± 2.86 whereas in CO_2 fertilized group it was 7.11±5. A low pH value in CO_2 fertilized group indicates the acidified condition of the cultures when compared to the control groups (Fig. 1a). The control and CO_2 fertilized group were strongly correlated $r(18) = 0.549$, $p<0.05$ and they are significant at 0.05 level (Fig. 1b).

Growth study of *Dunaliella salina*

Growth pattern on the acidification experiment of *D. salina* showed the average optical density value (0.047±0.180 OD) in control whereas in CO_2 fertilized it was reported as 0.072±0.076 OD. The growth curve in terms of cell density is shown in (Fig. 2a). The control and CO_2 fertilized group were strongly correlated $r(3) = 0.996$, $p<0.01$ and they are highly significant at 0.01 level (Fig. 2b).

Analysis of Chlorophyll 'a'

D. salina biomass in terms of chlorophyll 'a' and 'b' showed that the strains obtained the maximum amount of chlorophyll pigments in control group when compared with the CO_2 fertilized group. During the exponential phase of *D. salina* the chlorophyll 'a'content was found to be higher 0.066±0.077 (mg/ml) in control whereas in CO_2 fertilized it was lesser 0.152±0.122 (mg/ml) (Fig.3a). The control and CO_2 fertilized group were correlated $r(5) = 0.042$, $p>0.05$ and they are insignificant (Fig. 3b). Chlorophyll 'b' also showed similar results on the growth period,

higher chlorophyll 'b' (0.022± 0.003 mg/ml) was noticed in control group whereas in CO_2 fertilized it was 0.037± 0.017 (mg/ml) (Fig. 4a). The control and CO_2 fertilized group were correlated r (5) = 0.006, p>0.05 and they are insignificant (Fig. 4b).

The *D. salina* biomass in terms of phaeopigment was recorded maximum (40.84±7.13 mg/ml) in control group and minimum (49.93±23.12 mg/ml) in the CO_2 fertilized (Fig. 5a). The control and CO_2 fertilized group were strongly correlated r (5) = 0.969, p<0.05 and they are highly significant at 0.05 level (Fig.5b)

Biochemical analyses

The recorded total protein concentration in control and CO_2 fertilized was 1.44 (µg/ml) and 3.02 (µg/ml) respectively. In our observation, protein concentration was found to be very high in experimental group (CO_2 fertilized) when compared to the control (Fig. 6). Likewise, the total carbohydrate content also found to be high (0.41 µg/ml) in CO_2 fertilized experiment compared to control experiment (0.23 µg/ml) as shown in Fig.7. However, the lipid content was comparatively low (10%) in experimental group than control group (11%) (Fig. 8).

Discussion

Massive emissions of carbon dioxide into the atmosphere have an adverse impact on the chemical and biological processes in the ocean. There is a permanent exchange of gas between the air and the ocean. The dissolved carbon dioxide reacts to some extent to form carbonic acid. This reaction releases protons, which leads to acidification of the seawater. The present results showed the low pH values throught the experimental period which indicates the acidified condition due to CO_2 bubbling. The variation in pH can affect algal growth in a number of ways. It can change the distribution of carbon dioxide and carbon availability to species, alter the availability of trace metals and essential nutrients, and at extreme pH levels potentially cause direct physiological effects. As reported by Hinga (2002) responses of phytoplankton to reduced pH in a high CO_2 ocean are likely to be species specific, with potential "winners" and "losers". The sensitivity of a species reaction to CO_2 stressor and acidification cannot be defined alone by the simple formula: good acid-base regulation = high CO_2 tolerance. There are scientific studies that suggest this is not the case. It has been a controversial issue whether elevated CO_2 in seawater associated with atmospheric CO_2 rise would significantly promote phytoplankton productivity (Giordano *et al*., 2005).

In contrast, CO_2 tolerant species (such as *Dunaliella salina*) lack the state-transition response and display signs of oxidative damage under high - CO_2 conditions (Muradyan *et al.* 2004). CO_2 tolerance of *Dunaliella* sp. also has been examined and the species has been used in the industrial production of β-carotene (Graham and Wilcox, 2000). The use of carbon sequestration by micro algae is a major tool for reducing atmospheric concentrations of CO_2. Recent studies indicate that dominant phytoplankton species differ in their CO_2 requirement. Whereas some species preferably use CO_2 as a carbon source, others mainly draw their inorganic carbon from the large pool of HCO_3^- (e.g. Elzenga *et al*., 2000). Different algal species or growth conditions show differential preferences to CO_2 or HCO_3^- (Burkhardt *et al*., 2001; Nimer *et al*., 1997).

In our study, *D. salina* cultured has shown to produce high HCO_3^- concentration

throughout the experiment. The present results showed the acidified condition due to CO_2 bubbling. In the present results the cell density was rather high, doubled CO_2 increased the availability of DIC, thus the growth of *D. salina* might be stimulated in the stationary phase. The enhanced growth of the *D. salina* in control might be related to the increase of photosynthesis which has been shown with increased chlorophyll 'a' and 'b' contents.

The mechanistic implications of the effect of elevated CO_2 concentration on algal growth and productivity was previously studied (Hanagata *et al.*, 1992; Papazia., 2008; Hessen *et al*., 2008). The cells of *Dunaliella tertiolecta* grown under ordinary air (low-CO_2 cells) had a well developed pyrenoid with many more starch granules than those grown under air enriched with CO_2 (high-CO_2 cells). This could be the reason for high chlorophyll 'a' and 'b' was found presently in control group in comparison to the CO_2 fertilized group. The chloroplast was located close to the plasma membrane in low-CO_2 cells, while that in high- CO_2 cells it was located in the inner area of the cells. This implies that microalgae can possibly tolerate high concentration of CO_2 by adjusting their structural anatomy and redistribution of certain cellular organelles (Papazia *et al.*, 2008**)**.

Essa (1995) has described that the pH 9 was more suitable for protein production in *D. tertiolecta* and *D. salina*, while pH 6 was suitable for *D. bardawil*. The same author emphasized that pH 9 was more suitable than pH 6 for carbohydrate synthesis in *D. salina* and *D. parva*. Taha (2002) reported that both *D. tertiolecta* and *D. parva* need a lower pH for protein synthesis, the effect of different pH values on pigmentation of *D. bardawil* exhibited a similar trend to that of protein and carbohydrates, i.e., pH 7.5 also favored the accumulation of chlorophyll a, chlorophyll b, and carotenoids. *Dunaliella salina* responds to this decrease in CO_2 and increase in HCO3- at higher salinity by increasing the activity of a CO_2 concentrating mechanism (CCM) (Zenvirth and Kaplan 1981; Booth and Beardall 1991). This is accompanied by elevated levels of an external carbonic anhydrase (CA) (Booth and Beardall 1991). It has been routinely reported that enhancement of carbohydrate occurs in response to elevated CO_2 concentrations (Gordillo *et al*. 1999; Giordano 2001).

In *Chaetoceros cf. wighamii* carbon dioxide addition increased protein and lowered carbohydrates, but had no effect on lipid content (Sir Castro *et al*., 2005). This may be why CO_2 enrichment usually enhances photosynthesis, leading to an increase in soluble carbohydrate. It seems that the input of CO_2 would have stimulated the accumulation of carbohydrate, but not lipid. This is in accordance with taxonomic traits, since green algae are not rich in lipid (Machado and Lourenço, 2008). These data suggest an important effect of carbon availability to the production of carbohydrate by the chlorophycean microalgae. As a green alga, the accumulation of carbohydrate is a natural process under stimulating conditions (Graham *et al.* 2009). In the present study, the protein content was markedly increased with increasing CO_2 concentration, which in consistent with the report of Larrsson *et al*. (1985). Brown *et al.* (1997) noticed an increase (100%) in protein content when cultures were enriched with 1% carbon dioxide in many species in different groups of micro algae. In *Chaetoceros calcitrans* apparently directed the extra-assimilated carbon mainly to protein synthesis, indicating a positive effect on cell physiology. Probably,

the cells were investing the excess of carbon assimilated much more in protein synthesis and growth rather than lipids and carbohydrates as reserve substances in microalgae (Raghavan *et al*., 2008).

Conclusion

The present study was carried out to find out the response and effects of ocean acidification in the growth of green alga *Dunaliella salina* and its biochemical constituents viz. protein, lipid and carbohydrate. Ocean acidification is one of the most well-known ocean perturbations. The response of marine organisms to long-term increases in atmospheric CO_2 will depend on their ability to physiologically and morphologically adapt or acclimatize to expected changes in seawater chemistry. Utilizing naturally high CO_2 areas in order to reveal the evolutionary responses of marine algae to ocean acidification should be a next step for ocean acidification research, in order to predict the adaptive abilities to the current rapid acidification of our oceans. Adapting microalgae for the use of CO_2 sequestration also has the potential to produce useful byproducts, and could function multi-purposely. In addition, it is an environmentally friendly technology. Our study revealed that the *Dunaliella salina* considered being a promising candidate to overcome the ocean acidification since it showed the positive results on growth, pigments and biochemical's (protein and lipid) production in acidified conditions while carbohydrate contents was found diminishing in *D. salina*. Further it is inferred that ocean acidification impact is species dependent.

Acknowledgement

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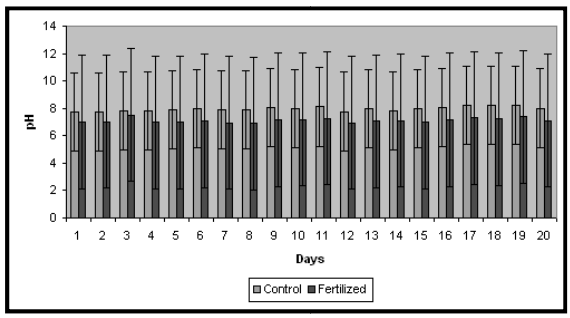

Fig. 1a: Measurement of pH in *Dunaliella salina*

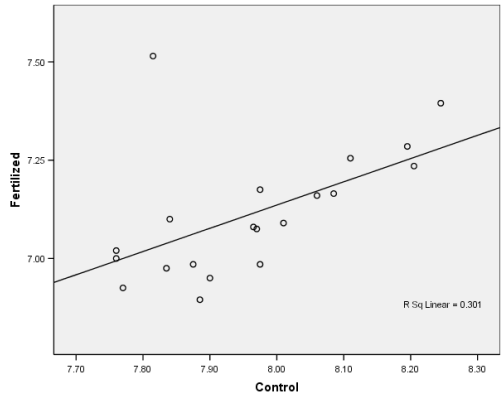

Fig. 1b: Correlation between pH of *Dunaliella salina* control and CO_2 fertilized

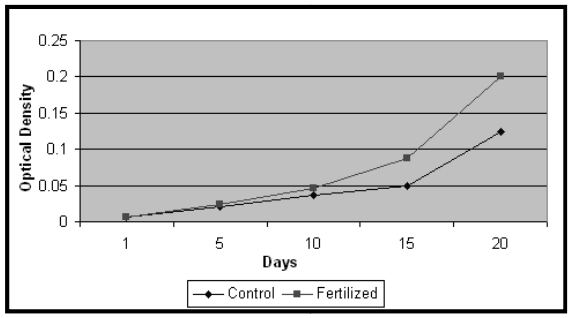

Fig. 2a: Growth rates of *Dunaliella salina*

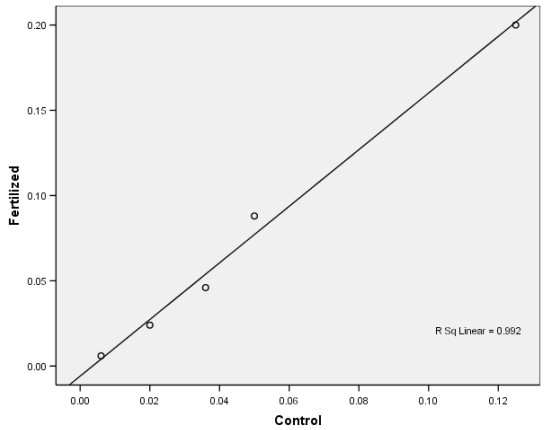

Fig. 2b: Correlation between growth rates of *Dunaliella salina* Control and CO_2 fertilized

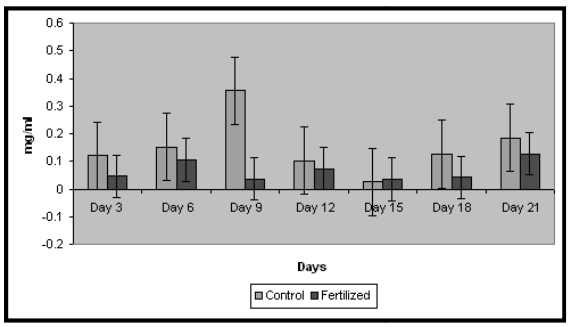

Fig. 3a: Estimation of chlorophyll 'a' in *Dunaliella salina*

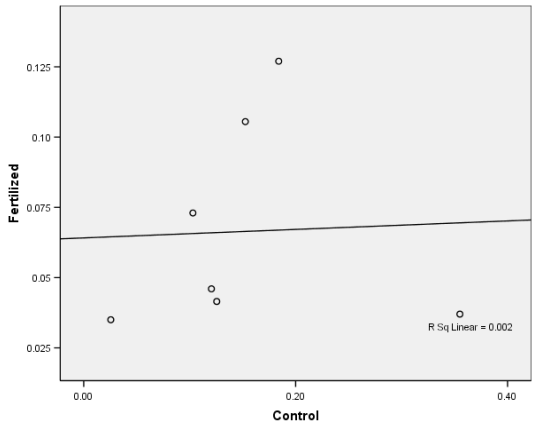

Fig. 3b: Correlation between chlorophyll 'a' in *Dunaliella salina* Control and CO_2 fertilized

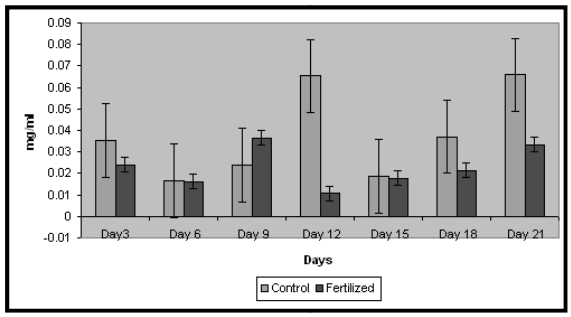

Fig. 4a: Estimation of chlorophyll 'b' in *Dunaliella salina*

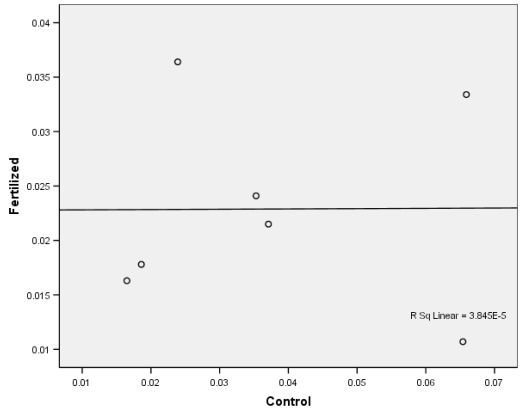

Fig 4b: Correlation between chlorophyll 'b' in *Dunaliella salina* Control and CO_2 fertilized

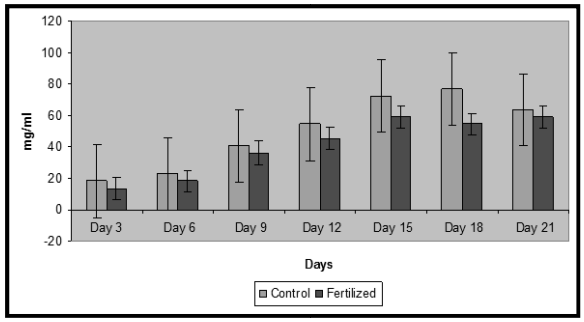

Fig. 5a: Estimation of phaeopigment production in *Dunaliella salina*

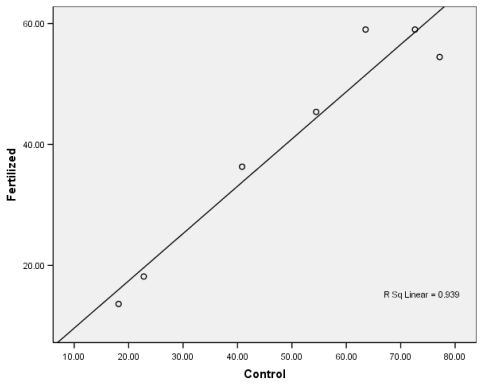

Fig. 5b: Correlation between phaeopigment production in *Dunaliella salina* control and CO_2 fertilized

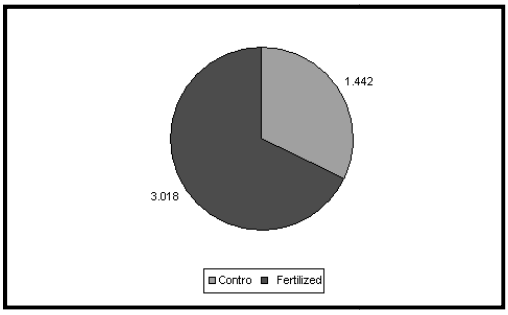

Fig. 6: Estimation of Total Protein in *Dunaliella salina* (µg/ml)

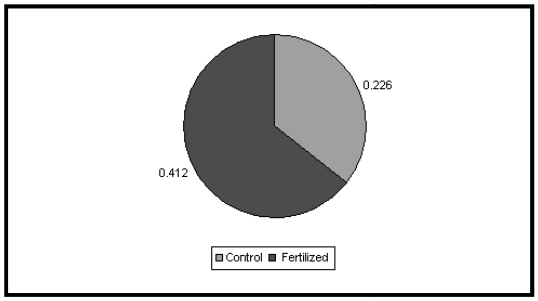

Fig. 7: Estimation of Total Carbohydrate in *Dunaliella salina* (µg/ml)

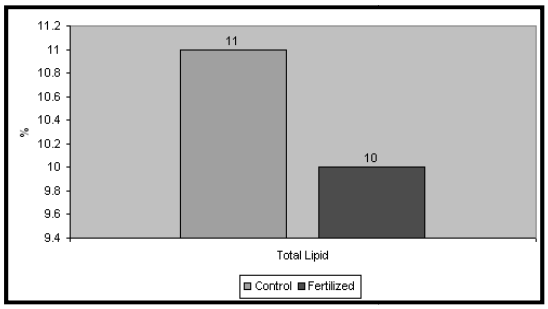

Fig. 8: Estimation of Total Lipid in *Dunaliella salina* (%)

References

Booth, W.A. and Beardall, J. 1991. Effects of salinity on inorganic carbon utilization and carbonic anhydrase activity in the halo tolerant alga *Dunaliella salina* (Chlorophyta). Phycologia. 30, 220-225.

Brown, M.R., Jeffrey, S.W., Volkman, J.K. and Dunstan, G.A. 1997. Nutritional properties of micro algae for mariculture. Aquaculture. 151, 315-331.

Brock, T. D., 1975. Salinity and the ecology of *Dunaliella* from Great Salt Lake. Microbiology. 89, 285–292.

Burkhardt, S., Amoroso, G., Riebesell, U. and Sultemeyer, D. 2001. CO_2 and HCO_3^- uptake in marine diatoms acclimated to different CO_2, concentrations. *Limnol Oceanogr*. 46, 1378-1391.

Dubois, M., Gilles, K.A., Hamilton, J.K., Rebers, P.A. and Smith, F. 1956. Colorimetric method for determination of sugars and related substances. *Anal. Chem*. 28: 350-356.

Elzenga, J.T., Prins, H.B. and Stefels, J. 2000. The role of extracellular carbonic anhydrase activity in inorganic carbon utilization of *Phaeocystis globosa* (Prymnesiophyceae): A comparison with other marine algae using the isotopic disequilibrium technique. *Limnol. Oceanogr*. 45, 372-380.

Folch, J., Lees, M. and Sloane-Stanley, G.H. 1957. A Simple Method for the Isolation and Purification of Total

Lipides from Animal Tissues, *J. Biol. Chem. 226*, 497–509.

Graham, L.E. and Wilcox, L.W. 2000. Algae, Prentice-Hall, Inc., Upper Saddle River NJ. pp 640.

Graham, L.E., Graham, J.E. and Wilcox, L.W., 2009. Algae. 2nd ed. Benjamin Cummings (Pearson). San Francisco. 720 p.

Hanagata, N., Takeuchi, T., Fukuju, Y., Barnes, D.J. and Karube, I. 1992. Tolerance of microalgae to high CO2 and high temperature. Photochemistry. 31(10), 3345-3348.

Hessen A.O. 2008. Excess Carbon in Aquatic Organisms and Ecosystems: Physiological, Ecological, and Evolutionary Implications. *Limnol Oceanogr*. 53 (3), 1685-1696.

Hinga, K.R. 2002. Effects of pH on costal marine phytoplankton. *Mar Ecol Prog Ser*. 238, 281-300.

Hileman, A. 1992. U.S. Urged to Change CO_2 Emissions Policy. *Chem Eng. News*, 70 (1), 16-22.

Ip, P.F. and Chen, F. 2005. Employment of reactive oxygen species to enhance astaxanthin formation in *Chlorella zofingiensis* in heterotrophic culture. *Process Biochem*. 40(11), 3491–3496.

Larsson, M., Larsson, C.M. and Guerrero, M. G. 1985. Photosynthetic nitrogen metabolism in high and low CO_2-adapted *Scenedesmus*. J. Exp. Bot. 36, 1373–1395.

Lowry, O.H., Rosebrough, N.J., Farr, A.L. and Randall, R.J. 1951. *J. Biol. Chem*. 193, 265-275.

Machado, R.R. and Lourenço, S.O. 2008. Proprieties nutritionist de microalgae usadas Como alimento de mollusc's bivalves: uma revisão. Ser. Livros, Museu Nacional, Rio de Janeiro, 30, 281-304.

Matsukawa, R., Hotta, M., Masuda, Y., Chihara, M. and Karube, I. 2000. Antioxidants from carbon dioxide fixing *Chlorella sorokiniana*. *J Applied Phycology*. 12, 263-267.

Muradyan, E.A., Klyachko-Gurvich, G.L., Tsoglin, L.N., Sergeyenko, T.V. and Pronina, N.A. 2004. Changes in lipid metabolism during adaptation of the *Dunaliella salina* photosynthetic apparatus to high CO_2 concentration. *Russ J Plant Physiol*. 51, 53–62

Nimer, N.A., Iglesias-Rodriguez, M.D. and Merrett, M.J. 1997. Bicarbonate utilization by marine phytoplankton species. *Journal of Phycology* 33, 625-631.

Oren, A. 2005. A century years of *Dunaliella* research: 1905-2005. Saline Systems,

Papazia, P., Makridisb, P., Divanachb and Kotza-basisa, K. 2008. Bioenergetic Changes in the Microalgal Photo-synthetic Apparatus by Extremely High CO_2 Concentrations Induce an Intense Biomass Production. *Physiologia Plantarum*. 132 (3), 338-349.

Raghavan Gireesh, Chanjaplackal Kesavan Haridevi and Cherukara Purushothaman Gopinathan, 2008. Growth and proximate composition of the *Chaetoceros calcitrans* f. pumilus under different temperature, salinity and carbon dioxide levels. Aqua Res. 39, 1053-1058.

Stepan, R.E., Shockey, T.A., Moe and Dorn, R. 2001. SUBTASK, 2.3-Carbon Dioxide Sequestering Using Microalgal Systems, U.S. Department of Energy, National Energy Technology Laboratory, United States of America.

Stumm, W. and Morgan, J.J. 1996. Aquatic Chemistry: Chemical Equilibria and Rates in Natural Waters, 3rd ed. Wiley, New York.

Schilipaulis, L. 1991. The Extensive Commercial Cultivation of *Dunaliella*. *Biores Technol*. DOI: 10.1016/0960-8524(91)90162.

Sir lei de Castro Araújo, Virginia Maria Tavano Garcia, 2005. Growth and biochemical composition of the diatom *Chaetoceros cf. wighamii* bright well under different temperature, salinity and carbon dioxide levels. I. Protein, Carbohydrates and Lipids". Aquaculture. 246 (1–4), 405–412.

Taha, H.M. 2002. Comparative physiological and chemo taxonomical studies of some species of *Dunaliella* (Volvocales). Ph.D. Thesis. Fac. of Sci. Alex. Univ. Alex. Egypt.

Walne, P.R. 1966. Large scale culture of larvae *Ostrea edulis* L. Fish Invest., II 25, 1–52.

Zenvirth, D. and Kaplan, A. 1981. Uptake and efflux of inorganic carbon in *Dunaliella salina*, Planta. 152, 8-12.